generally require students to reflect on their research as part of their project reports to be able to evaluate their learning from the research process. In Section 1.5 we talked about keeping a reflective diary and provided a checklist to help you to do this (Box 1.4).

Reflection may occur in a number of ways. It may occur during an event, so that you reflect on your approach while you are conducting an activity. This type of reflection may occur, for example, while you are interviewing or observing. Reflection may also occur after an activity has taken place so that you reflect on what occurred and how you might be able to do better next time. A more fundamental type of reflection, known as reflexivity, involves you in monitoring and reflecting on all aspects of the research project from initial ideas to submission of the project report (Section 2.1). It includes examining your reactions to what is being researched, the nature of your relationship with those who take part in the research and evaluating the way in which you interpret data to construct knowledge (Haynes 2012). Given its interpretivist nature, Finlay (2002: 211) says that reflexivity is 'now the defining feature of qualitative research'. Your reactions, your interactions with those taking part and your attitudes and beliefs may each impact on your interpretation of the data that are shared with you. Engaging in forms of reflexivity may enable you to develop greater insights as you explore and analyse these data. Developing a reflexive focus in your reflective diary may therefore prove to be a valuable aid to further your research (Section 1.5).

13.6 Thematic Analysis

Introduction

We start by outlining **Thematic Analysis** as this is often thought of as a general approach to analysing qualitative data. Braun and Clarke (2006: 78) refer to Thematic Analysis as a 'foundational method for qualitative analysis'. The process of thematic analysis is found in other approaches to qualitative analysis, albeit in more particularised ways, as we outline in the following sections. The essential purpose of this approach is to search for themes, or patterns, that occur across a data set (such as a series of interviews, observations, documents, diaries or websites being analysed). Thematic Analysis involves a researcher coding her or his qualitative data to identify themes or patterns for further analysis, related to his or her research question. We discuss procedures to undertake analysis in the next part of this section.

Thematic Analysis offers a systematic yet flexible and accessible approach to analyse qualitative data (Braun and Clarke 2006). It is systematic as it provides an orderly and logical way to analyse qualitative data. In this way, Thematic Analysis can be used to analyse large qualitative data sets, as well as smaller ones, leading to rich descriptions, explanations and theorising. Thematic Analysis can be used to help you:

1. comprehend often large and disparate amounts of qualitative data;
2. integrate related data drawn from different transcripts and notes;
3. identify key themes or patterns from a data set for further exploration;
4. produce a thematic description of these data; and/or
5. develop and test explanations and theories based on apparent thematic patterns or relationships;
6. draw and verify conclusions.

Thematic Analysis is flexible as it is not tied to a particular research philosophy. You may use Thematic Analysis irrespective of whether you are adopting an objectivist or subjectivist position (Chapter 4). Your assumptions will, however, affect how you use it to interpret your data (which is why you should be explicit about your philosophical assumptions and remain reflexive through your research project). As a realist you may use Thematic Analysis to seek to understand factors underpinning human attitudes and actions. Alternatively, as an interpretivist you may use it to explore different interpretations of a phenomenon. The reason why you may use Thematic Analysis irrespective of your philosophy relates to its development as a standalone analytical technique or process, rather than being part of a theoretically mounted methodological approach.

For the same reason, Thematic Analysis may be used irrespective of whether you adopt a deductive, inductive or abductive approach. In a deductive approach, the themes you wish to examine would be linked to existing theory. Your research question is also more likely to be firmly established and this and your research objectives may be used to derive themes to examine in your data. This may lead you to focus on parts of your data set rather than seek to analyse it all in an undiscriminating way. In an inductive approach, themes will be derived from the data. You will search for themes to explore related to your research interest but will not impose a framework of themes to examine your data set based on existing theory. Depending on which themes you decide to explore in an inductive approach, you may also modify your research question. Initially you will be likely to explore the whole data set looking for the occurrence and reoccurrence of themes. You may also use an abductive approach, commencing analysis with theoretically-derived themes which you then modify or add to as you explore your data set.

The nature and flexibility of Thematic Analysis mean that it is fairly straightforward to use in comparison to some of the techniques discussed later. Where you use Thematic Analysis, your energy can be invested in making sure your analysis is rigorous, rather than spending lots of time checking you are applying a more particularised approach to qualitative analysis according to strict rules advocated for its use. We now outline the procedure used in Thematic Analysis.

Procedure

The procedure outlined here provides a set of guidelines to undertake Thematic Analysis. In practice, this procedure does not occur in a simple linear progression. Instead it is likely to occur in a concurrent and recursive fashion, involving you analysing data as you collect them and going back over earlier data and analysis as you refine the way in which you code and categorise newly collected data and search for analytical themes.

The procedure outlined here involves four elements: becoming familiar with your data; coding your data; searching for themes and recognising relationships; refining themes and testing propositions. We now consider each of these.

Becoming familiar with your data

You will start to become familiar with your data as you produce transcripts of the interviews or observations you conduct, or as you read through documents or diaries or review visual images. The act of transcribing a data item yourself, although laborious, allows you to develop familiarity. This should also prompt you to generate summaries, self-memos or entries in your notebook that aid your analysis.

Familiarisation with your data involves a process of immersion that continues throughout your research project. You will need to read and re-read your data during your analysis. You will be interested to look for meanings, recurring themes and patterns in your data. Without familiarity, you will not be able to engage in the analytical procedures that follow. Producing transcripts and data familiarisation are therefore important elements in analysing data.

Coding your data

Coding is used to categorise data with similar meanings. **Coding** involves labelling each unit of data within a data item (such as a transcript or document) with a code that symbolises or summarises that extract's meaning. Your purpose in undertaking this process is to make each piece of data in which you are interested accessible for further analysis (Boxes 13.5 and 13.6). Qualitative data sets are frequently large and their content complex. A qualitative data set may include references to actions, behaviours, beliefs, conditions, events, ideas, interactions, outcomes, policies, relationships, strategies, etc. Without coding these data you may struggle to comprehend all of the meanings in your data in which you are interested. Coding is therefore an important means to manage your data so that you can rearrange and retrieve them under relevant codes. This process effectively involves fragmenting your original data items and regrouping units of data with similar meanings together to be able to examine them in relation to other groups of similar units of data. The act of coding your data is therefore a vital element in data analysis, as you will see.

A **code** is a single word or a short phrase, which may also be abbreviated in use (Boxes 13.5 and 13.6). A coded extract of data is referred to as a unit of data. A **unit of data** may be a number of words, a line of a transcript, a sentence, a number of sentences, a complete paragraph, other chunk of textual data, or visual image that is summed up by a particular code (Boxes 13.5 and 13.6). The exact size of a unit of data will be determined by its meaning. Some units of data will overlap and some will be coded using more than one code (Box 13.5).

If you think that a new piece of data has a similar meaning to a previously coded unit of data, it should be labelled with the same code. If you think that a new piece of data does not have a similar meaning to a previously coded unit of data, you will need to devise a new code for it. Throughout the process of coding it will be important to keep a list of codes you are using and a working definition for each, to ensure consistency.

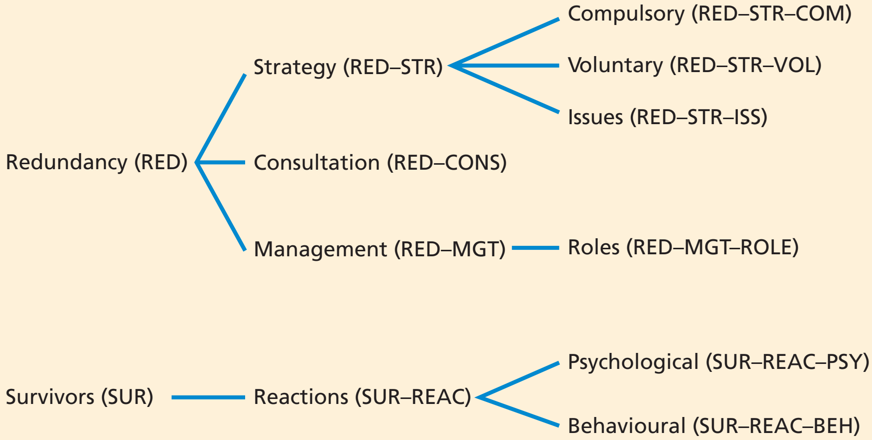
At this point you may be asking two questions. How much of my data should I code – all or only some of it? Where should my codes come from? Both of these questions are related to your research approach and also to your research question – whether you are setting out to use an inductive or deductive approach and how well you have defined your research question. We now answer each of these questions in turn.

How much of your data you code will depend upon your research approach and research question. Where you use a purely inductive approach you will be likely to code all of it, as you explore all possible meanings in your data to guide the direction of your research. This search for meanings may also lead to finely detailed coding, where you find yourself coding smaller segments or units of data to capture every possible nuance. Where you use a purely deductive approach, you will commence with a framework of codes derived from prior conceptual or theoretical work. In this case you will commence coding by applying these prior codes to your data. Using a purely inductive or deductive approach may be problematic. A purely inductive approach may mean that you spend a great deal of time coding every possible unit of data before you decide on a particular research focus. Using a purely inductive approach is appropriate for a very exploratory study but you would need to ensure that you have ample time to conduct it, perhaps related to a major research project. Where you use an inductive approach and have defined a research question, you should be able to use this question to help select which data to code. In this case, while all of your data may be potentially interesting, your research question will help you focus on which data to code. Using a purely deductive approach may lead you to conclude that your list of prior codes is inadequate and that you need to devise other codes in order to be able to code your data adequately to begin to answer your research question and address your research objectives.

Box 13.5
Focus on student research

Interview extract with categories attached

Adrian's research project was concerned with how human resource management professionals managed a downsizing process in their own organisations. He derived initial codes from existing theory in the academic literature and attached them to appropriate units of data in each transcript. His initial categories were hierarchical; the codes he used being shown in brackets:



These were then attached to the interview transcript, using sentences as units of data. Like our jigsaw example at the start of this chapter, those units of data that were coded with more than one category suggested interrelationships:

RED–CONS	27MM The first stage is to find out what particular employees
	want for themselves and how they want this to happen. Staff are
	seen by their line manager and/or a member of personnel.
RED–MGT–ROLE	Employees might want to talk to someone from personnel rather
	than talk with their line manager – well, you know, for obvious
	reasons, at least as they see it – and this would be acceptable to the
RED–STR–VOL	organisation. This meeting provides them with the opportunity to
	opt for voluntary redundancy. We do not categorise employees
RED–STR–ISS	into anything like core or non-core, although we will tell a group
RED–CONS	of employees something like ‘there are four of you in this
	particular function and we only need two of you, so you think
RED–CONS	about what should happen’. Sometimes when we attempt to give
	employees a choice about who might leave, they actually ask us to
	make the choice. This is one such situation where a compulsory
RED–STR–COM	selection will occur. We prefer to avoid this compulsory selection
SUR–REAC–PSY	because of the impact on those who survive – negative feelings,
	guilt and so on.

This discussion indicates where your codes may come from. There are three main sources of codes which, dependent on your approach to theory development can be used on their own or in combination. Codes may be:

- actual terms used by your participants, recorded in your data. These are often referred to as 'in vivo' codes;
- labels you develop from your data;
- derived from existing theory and the literature. These are often referred to as 'a priori' codes.

These sources of codes are shown in Figure 13.1 to illustrate their relationship.

Coding is a simple but versatile and valuable tool. The process of coding allows you to link units of data that refer to the same aspect or meaning, or to link aspects or meanings that you want to compare and contrast. It allows you to rearrange your original data into groupings for the next stage of analysis. Any unit of data may be coded with as many different codes as you think is appropriate, creating a web of connections to aid your analysis (Boxes 13.5 and 13.6). It is often important to understand the context of the data you are analysing. Where it is important to include some contextual background, you can code larger units of data such as whole paragraphs, as opposed to smaller units such as a few words or single sentences. You should also note that codes may be referred to as categories: these terms are sometimes used interchangeably and sometimes to refer to different aspects of the analytical process – see the next sub-section.

Your approach to coding will be guided by the purpose of your research as expressed through your research question and objectives. Another researcher with different objectives to you may derive different codes from the same data. You will be likely to develop new codes as you conduct more interviews or observations and expand your data set. You will also be likely to gain new insights from existing codes that suggest new ones during the process of analysis. This will require you to re-read all of your earlier data transcripts to re-code them according to your current list of codes. This process is termed constant comparison and is undertaken to ensure consistency in the way you code and analyse your data set.

Your codes will show the occurrence or non-occurrence of a phenomenon and the strength of opinion in some instances. Some codes may attract large numbers of units of data. Some of these may prove to be too broad for further analysis without being subdivided. For example, Adrian undertook a research project where some codes had large amounts of data attached to them, while others attracted relatively small amounts of data. This led to the large codes being subdivided into further codes, which was helpful in pursuing the analysis (Box 13.5). Codes attracting small numbers of units of data may be merged with similar ones or retained until later in the process of analysis in case they prove to be more important than they appear initially.

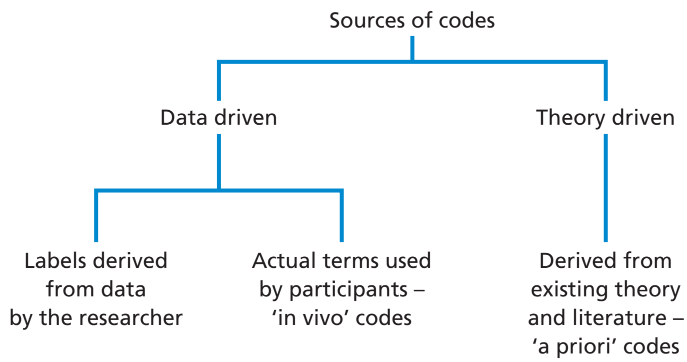


Figure 13.1 Sources and types of code

Box 13.6 Focus on management research

Developing, revising and applying codes to analyse data

In an article published in the *Journal of Occupational and Organizational Psychology* (2016: 634), McConville, Arnold and Smith analysed "qualitative data to explore how employees perceive the relationships between employee share ownership scheme participation, their attitudes and behaviours at work, and their feelings of psychological ownership." They used Thematic Analysis to analyse data collected from 37 semi-structured interviews which were conducted in nine companies with participants in employee share ownership schemes.

Initial codes to analyse these data were developed deductively from relevant theory and inductively during the process of data familiarisation following its collection. Interviews were audio-recorded and transcribed, aiding the process of data familiarisation. Following some revision to their initial set of codes, each data transcript was coded holistically using this revised set of codes. This led them to consider the units of data allocated to these codes in greater detail, which in turn led to some of these codes being revised further and new codes being added, with data being recoded as a consequence of these changes. Their introduction of new codes also suggests that these started to become more hierarchical since they refer to introducing sub-codes to recognise the relationship between an observed behaviour and reasons given for this by participants. They provide an example of these new codes and sub-codes.

They introduced a new code, 'CIT', to label incidents of organisational citizenship behaviours (OCB) which were evident in the data. The introduction of this new code led them to examine the reasons for incidents of OCB. One reason was linked to participations in the employee share ownership scheme. To recognise the relationship between incidents of OCB and employee share ownership participations they therefore developed a sub-code called 'CIT/ESO'.

These examples illustrate that as data analysis proceeded it became a recursive activity in its own right: codes were applied to data, leading to units of data being analysed further in relation to the application of these codes, leading to the revision of some codes and the development of others to refine analysis and recognise relationships between different codes and the units of data attached to these.

You may use CAQDAS to help you to code your data (Section 13.14) or you may use a manual approach. Where you use the second approach, you can label a unit of data with the appropriate code (or codes) in the margin of your transcript or set of notes (Box 13.6). This may then be copied, cut up and stuck onto a data card, or otherwise transferred and filed so that you end up with piles of related units of data. When doing this, it is essential to label each unit of data carefully so that you know its precise source (Section 13.4). An alternative is to index codes by recording precisely where they occur in your transcripts or notes (e.g. interview 7, page 2, line 16) on cards headed with particular codes. Undertaking this stage of the analytic process means that you are engaging in a selective process, guided by the aim of your research and your research objectives, which has the effect of reducing and rearranging your data into a more manageable and comprehensible form.

One way of achieving this reduction and rearrangement of your data, depending on its suitability, is to use one or more of the analytical techniques described by Miles et al. (2014). These are considered later in Section 13.13.

Searching for themes and recognising relationships

This is seen as a distinct stage of analysis that follows coding, although in practice you will be searching for themes, patterns and relationships in your data as you collect and code them. Producing progress summaries, transcript summaries, document summaries, self-memos and/or entries in your research notebook and reflective diary will help you to record your ideas about possible themes, patterns and relationships in your data.

The search for themes fully begins when you have coded all of your data set. At this point you will have a long list of codes that you have created to make sense of and draw meaning from your data. Advice about the number of codes you might be working with at this point varies considerably. Some advice refers to working with up to 30 codes. Other advice refers to creating as many codes as you require to interpret every relevant meaning in your data. This may mean creating up to a couple of hundred codes, or possibly more. Our view is that data should not be forced into a particular number of codes. The number of codes you create will be related to the meanings you wish to explore in your data set, the nature of your research approach and the focus of your research question. However, where you find yourself creating very large numbers of codes you will need to evaluate whether your coding is too detailed or whether you are trying to analyse too much data for your project. Always refer back to your research question, research aim and research objectives to focus your approach to data analysis.

This stage of analysis involves you searching for patterns and relationships in your long list of codes to create a short list of themes that relate to your research question. A **theme** is a broad category incorporating several codes that appear to be related to one another and which indicates an idea that is important to your research question. A theme may also be a single code which indicates an idea that assumes general importance to your research question and is therefore elevated to become a theme. Searching for themes is part of the overall process of condensing your raw data, firstly by coding them and then grouping these coded data into analytic categories.

In some discussions, terms like codes and themes are used interchangeably and this can lead to confusion (Ritchie et al. 2014). We are sometimes asked about the difference between a code and a theme. One way we have found it helpful to explain this is to say that data are organised by coding them while codes are organised by drawing them together as themes. This distinction reflects what we outline here. Sometimes you will also see the term 'thematic code': this is simply an alternative way to refer to the type of coding we describe here, where your coding leads to identification of themes as opposed to coding for actions as advocated by Charmaz (2014) in constructivist grounded theory, which we discuss in Section 13.9.

Searching for themes involves you making judgements about your data and immersing yourself in them. You will be looking to see how the codes you have created might fit together to allow you to further your analysis. As you search these codes, some initial questions you may ask include:

- What are the key concepts in these codes?
- What, if anything, seems to be recurring in these codes?
- What seems to be important, whether it recurs often or not?
- What patterns and/or trends are evident in the coded data?
- Which codes appear to be related?
- How do a particular set of codes appear to be related?

As you start to decide on themes to analyse your data further, some additional questions you may ask include:

- What is the essence of each apparent theme?
- How might themes be related to each other?
- Which themes appear to be main themes and which appear to be sub-themes (related to a main theme)? There may also be third level themes evident in your analysis.

- How may the relationship between themes be represented (as a hierarchy or a network) to produce a thematic map?
- Is there an overarching theme (or more than one) that unites your analysis?

You should not expect this process to be unproblematic. In attempting to achieve a thorough understanding of your data set, some further questions you may ask include:

- How well does this initial thematic map represent the relationships between themes?
- Which themes, if any, do not fit within this thematic representation?
- Does the way the data have been coded need to be revised; if so which data and how?
- Which themes need to be refined, discarded or newly introduced?
- How may the thematic representation be modified to represent my data better?

In the first set of questions, you begin to decide on themes to further your analysis. In the second set of questions, you begin to define your themes and the relationships between them (Box 13.5). Some themes will become main themes; some may become secondary-level themes, linked to a main theme; yet others may be tertiary-level themes, linked to a secondary-level theme. In the third set of questions, you evaluate your themes and the relationships between them. This will mean refining your themes and testing proposed relationships, as we discuss further in the next sub-section.

Refining themes and testing propositions

Refining themes and the relationships between them is likely to be an important part of your analytical process. The themes that you devise need to be part of a coherent set so that they provide you with a well-structured analytical framework to pursue your analysis. As you develop themes you should reorganise your coded data extracts under the relevant theme or sub-theme. This will help you to evaluate whether these coded data are meaningful to one another within their theme and whether (and how) themes are meaningful in relation to one another and in relation to your data set. This is likely to be a developmental process, as you re-read and reorganise your data. As you continue to examine your data set, the codes you have used and the themes you devise to organise your coded data to answer your research question, you will be likely to refine these themes.

You may decide that some of your initial themes should be combined to make a new theme while others should be separated into different themes. You may also decide that some of your initial themes should be discarded. Your decisions to make these changes will be based on re-reading the coded data that you have reorganised under each relevant theme. By reading the coded data attached to a possible theme, you will be able to evaluate whether these data support the continuation of the theme, or whether there is insufficient data to sustain it. This will allow you to decide whether these data are too dissimilar so that these should be separated into more than one theme. It will also allow you to decide whether two or more themes contain similar meanings and so should be collapsed into a single theme. As you refine your themes in this way you will also be able to revise the relationships between them.

As you seek to reveal patterns within your data and to recognise relationships between themes, you will be able to develop testable propositions (Box 13.7). The appearance of an apparent relationship or connection between themes will need to be tested if you are to be able to conclude that there is an actual relationship. However, while this is sometimes referred to as 'testing a hypothesis', it is not the same as the statistical hypothesis or significance testing we discuss in relation to quantitative analysis in Section 12.5.

It is important to test the propositions that emerge inductively from the data by seeking alternative explanations and negative examples that do not conform to the pattern or relationship being tested. Alternative explanations frequently exist, and only by testing

Box 13.7 Focus on student research

Research propositions

During the process of qualitative data analysis, a student evaluating the use of online retailing formulated the following proposition:

Customers' willingness to trust specific online retailers depends on their previous customers' reviews.

A student exploring mortgage borrowers' decision making drew up this proposition:

Potential mortgage borrowers' choice of lending institution is strongly affected by the level of customer service that they receive during the initial inquiry stage.

Another student investigating cause-related marketing formulated the following proposition:

Companies engaging in cause-related marketing are motivated principally by altruism.

A relationship is evident in each of these propositions. Each was tested using the data that had been collected.

the propositions that you identify will you be able to move towards formulating valid conclusions and an explanatory theory, even a simple one (Miles et al. 2014). Dey (1993: 48) points out that 'the association of one variable with another is not sufficient ground for inferring a causal or any other connection between them'. The existence of an intervening variable may offer a more valid explanation of an association that is apparent in your data (Box 13.8).

By rigorously testing your propositions against your data, looking for alternative explanations and seeking to explain why negative cases occur, you will be able to move towards the development of valid/credible and well-grounded conclusions. The validity/credibility of your conclusions needs to be verified by their ability to withstand alternative explanations and the nature of negative cases. **Negative cases** are those that do not support your

Box 13.8 Focus on student research

The impact of an intervening variable

Kevin's research project involved looking at the use of subcontractors by an organisation. A relationship appeared to emerge between the total value of contracts a particular subcontractor had been awarded and the size of that contractor in terms of number of employees; in particular, those contractors with larger numbers of employees had a larger total value of contracts. This could have led Kevin to conclude that the value of work undertaken by a particular subcontractor was related to that organisation's size and that, in particular, the organisation tended to use subcontractors with large numbers of employees.

Reality was not so simple. The organisation had originally used over 2,500 subcontractors but had found this exceedingly difficult to manage. To address this issue the organisation had introduced a system of preferred contractors. All 2,500 subcontractors had been graded according to the quality of their work, with those whose work had been consistently of high quality being awarded preferred contractor status. This meant that they were invited by the organisation Kevin was researching to tender for all relevant contracts. The intervening variable was therefore the introduction of preferred contractor status dependent upon the quality of work previously undertaken. The fact that the majority of these subcontractors also had relatively large numbers of employees was not the reason why the organisation had awarded them contracts.

explanations and the induction of your grounded theory. Finding cases that do not fit with your analysis should be seen positively as these will help to refine your explanations and direct the selection of further cases to collect and analyse data.

This will help you to avoid interpretations that prove to be unreliable because you only notice evidence that supports your own opinions. It relates to our discussion of reflexivity in Section 13.5. As a researcher you need to recognise your own attitudes and beliefs about the topic being researched, perhaps by writing about these to make them explicit, in order to understand how this affects your judgement about what the research data might mean and to gain greater insights while analysing these data. Brinkmann and Kvale (2015: 278) refer to this process as seeking to achieve 'reflexive objectivity'.

Evaluation

Thematic Analysis offers a systematic approach to qualitative data analysis that is accessible and flexible. Compared to some qualitative approaches, it is not overly prescriptive about the application of its analytical procedures. As a generic approach to qualitative data analysis it is suitable to use with several qualitative research strategies, where you are not following a named version of a strategy that prescribes precise analytic procedures, as in Grounded Theory Method.

Thematic Analysis may be used to induce theory in a similar way to Grounded Theory Method, but without following its prescribed approach to coding (Section 13.9). It may also be used to produce descriptive or explanatory accounts that fall short of generating a grounded theory.

Thematic Analysis is more adaptable, so that if the research strategy you are using requires you to search for particular themes you may consider using it. The process of searching for themes is common to other analytical approaches, as we consider in the following sections of this chapter.

Thematic Analysis may also be used in relation to deductive and inductive research approaches. Using a purely deductive or inductive research approach may be problematic, affecting the scope of the analysis. Thematic Analysis allows the researcher to move between these approaches.

13.7 Template Analysis

Introduction

Template Analysis is a specific type of thematic analysis, with a few key differences to Thematic Analysis as outlined in Section 13.6. In Thematic Analysis all data items (transcripts or other text) are coded first before the search for interpretive themes fully begins (Section 13.6). This is to avoid early thematic interpretation prematurely shaping or skewing the direction of the research in this emergent approach. This concern is also recognised in Template Analysis, and while only a proportion of the data items are coded before developing an initial coding structure and interpretive themes, known as a coding template, these data items are chosen for their representativeness or heterogeneity to try to overcome it (King and Brookes 2017).

This **coding template** is a hierarchical representation of themes and codes (Section 13.6) and is used as the central analytical tool in Template Analysis. A researcher using Template Analysis will start by coding a sufficient part of their data to develop an initial coding template (Box 13.9). This may mean coding a small number of interview or